🌱 ENVIRONMENT AND ITS CONSERVATION

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1. Introduction to Environment

The environment refers to everything that surrounds an organism and influences its life. It includes both living and non-living components that interact with each other to form a stable natural system. Every plant, animal and human depends on the environment for survival, food, shelter, air and water.

The relationship between organisms and their surroundings is continuous and dynamic. Any change in one part of the environment affects other parts. For example, removal of forests affects rainfall, soil fertility and wildlife population.

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2. Components of the Environment

The environment is broadly divided into two main components: biotic and abiotic.

2.1 Biotic Components

Biotic components include all living organisms present in an environment. These are further classified into producers, consumers and decomposers.

Producers:
Producers are green plants that prepare their own food using sunlight, carbon dioxide and water through the process of photosynthesis. They form the base of all food chains.

Consumers:
Consumers depend directly or indirectly on producers for food. They are divided into herbivores, carnivores and omnivores.

Decomposers:
Decomposers such as bacteria and fungi break down dead plants and animals into simpler substances and return nutrients to the soil.

2.2 Abiotic Components

Abiotic components include all non-living factors such as air, water, soil, sunlight, temperature and minerals. These components determine the type of organisms that can survive in a particular area.

For example, polar regions have very low temperatures, so only cold-adapted animals like polar bears can survive there.

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3. Ecosystem

An ecosystem is a functional unit of the environment where living organisms interact with each other and with the physical environment. Examples include forests, ponds, rivers, deserts and oceans.

Each ecosystem contains producers, consumers and decomposers working together to maintain balance.

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4. Natural Cycles

Nature maintains balance through various biogeochemical cycles.

4.1 Water Cycle

Water continuously circulates through evaporation, condensation and precipitation. This cycle ensures the availability of freshwater for all living organisms.

4.2 Carbon Cycle

Carbon moves between the atmosphere, plants, animals and soil. Plants absorb carbon dioxide during photosynthesis, while animals release it during respiration.

4.3 Nitrogen Cycle

Nitrogen is essential for making proteins. Certain bacteria convert nitrogen into usable forms for plants.

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5. Human Impact on Environment

Human activities have greatly affected the environment. Deforestation, industrial growth, urbanization and overuse of natural resources have disturbed the natural balance.

Pollution from vehicles and factories increases harmful gases in the atmosphere. Excessive use of fertilizers pollutes soil and water bodies.

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6. Environmental Problems

Environmental problems include air pollution, water pollution, soil pollution, global warming and climate change.

These problems cause health issues, affect agriculture and reduce biodiversity.

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7. Conservation of Environment

Conservation refers to the protection and wise use of natural resources.

Important measures include planting trees, conserving wildlife, reducing waste and promoting renewable energy.

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8. Sustainable Development

Sustainable development aims to meet present needs without harming future generations. It promotes responsible use of resources.